Furthermore,

$$B(\beta,\gamma) = -k_i k_{i'}$$

since Γ_1, Γ_2 are only connected at ϵ. But this has to be a nonnegative number, since $k_i > 0$ and $k_{i'} \leq 0$. This yields

$$B(\alpha,\alpha) = B(\beta+\gamma,\beta+\gamma) = \underbrace{B(\beta,\beta)}_{\geq 2} + 2\underbrace{B(\beta,\gamma)}_{\geq 0} + \underbrace{B(\gamma,\gamma)}_{\geq 2} \geq 4.$$

But this is a contradiction, since α was assumed to be a root. □

Definition 6.4.7. We call a root $\alpha = \sum_i k_i\alpha_i$ a positive root if all $k_i \geq 0$. A root for which $k_i \leq 0$ for all i is called a negative root.

Remark 6.4.8. Lemma 6.4.6 states that every root is either positive or negative.

Example 6.4.9. (1) Let Γ be of the type A_{N-1}. Then the lattice $L = \mathbb{Z}^{N-1}$ can be realized as a subgroup of the lattice $\mathbb{Z}^N$ by letting $L \subseteq \mathbb{Z}^N$ be the subgroup of all vectors $(x_1,\ldots,x_N)$ such that

$$\sum_i x_i = 0.$$

The vectors

$$\begin{aligned}
\alpha_1 &= (1,-1,0,\ldots,0),\\
\alpha_2 &= (0,1,-1,0,\ldots,0),\\
&\vdots\\
\alpha_{N-1} &= (0,\ldots,0,1,-1)
\end{aligned}$$

naturally form a basis of L. Furthermore, the standard inner product

$$(x,y) = \sum x_i y_i$$

on $\mathbb{Z}^N$ restricts to the inner product B given by Γ on L, since it takes the same values on the basis vectors:

$$(\alpha_i,\alpha_i) = 2,$$

$$(\alpha_i,\alpha_j) = \begin{cases} -1, & i,j \text{ are adjacent},\\ 0, & \text{otherwise}.\end{cases}$$

This means that vectors of the form

$$(0,\ldots,0,1,0,\ldots,0,-1,0,\ldots,0) = \alpha_i + \alpha_{i+1} + \cdots + \alpha_{j-1}$$